

Exercise 1

Advanced Methods for Regression and Classification

Kyzer Gerez

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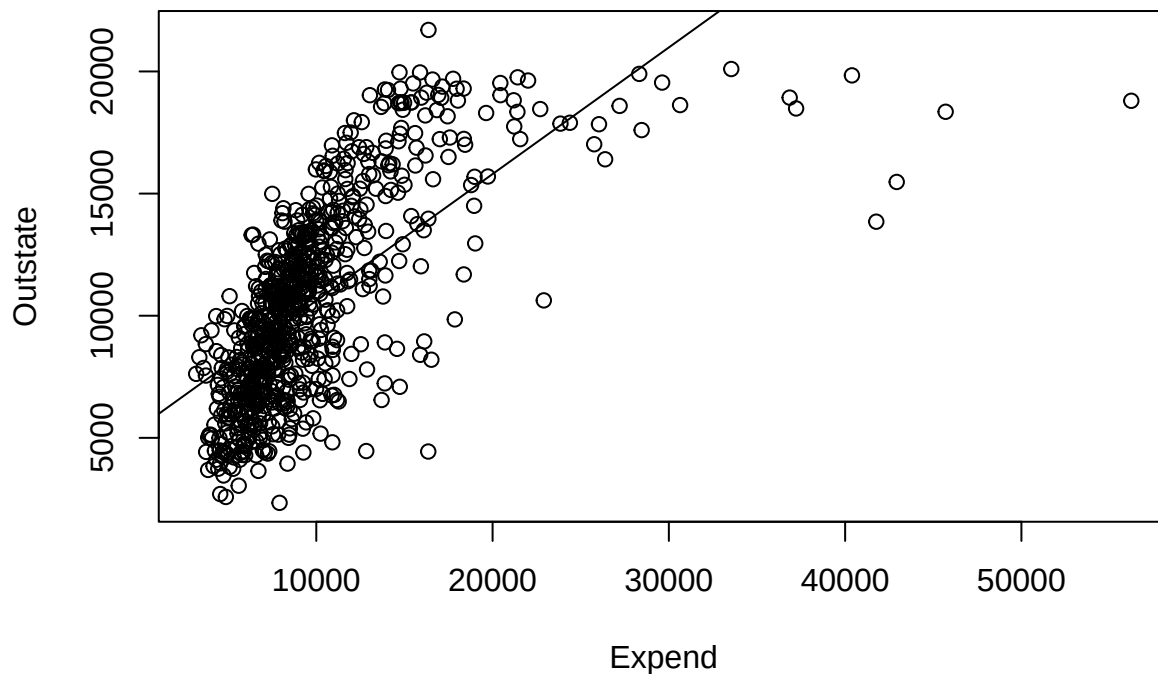
```
data(College, package="ISLR")  
sum(is.na(College)) # original data frame has no observations with missing data
```

```
## [1] 0
```

1. Predict the response Outstate by using only Expend as predictor. Plot the data, and visualize the regression line (abline()). What do you conclude?

```
College_outstate_lm <- lm(Outstate ~ Expend, College)
```

```
plot(x = College$Expend,  
     y = College$Outstate,  
     xlab = "Expend",  
     ylab = "Outstate")  
abline(College_outstate_lm)
```



Outliers in the Expend dimension are influencing the regression line to take on a more narrow slope.

2. It seems that the model is somewhat biased. Can you come up with a “more appropriate” model that better follows the visually visible linear trend? Just ideas: transformation, higher order terms, outlier handling.

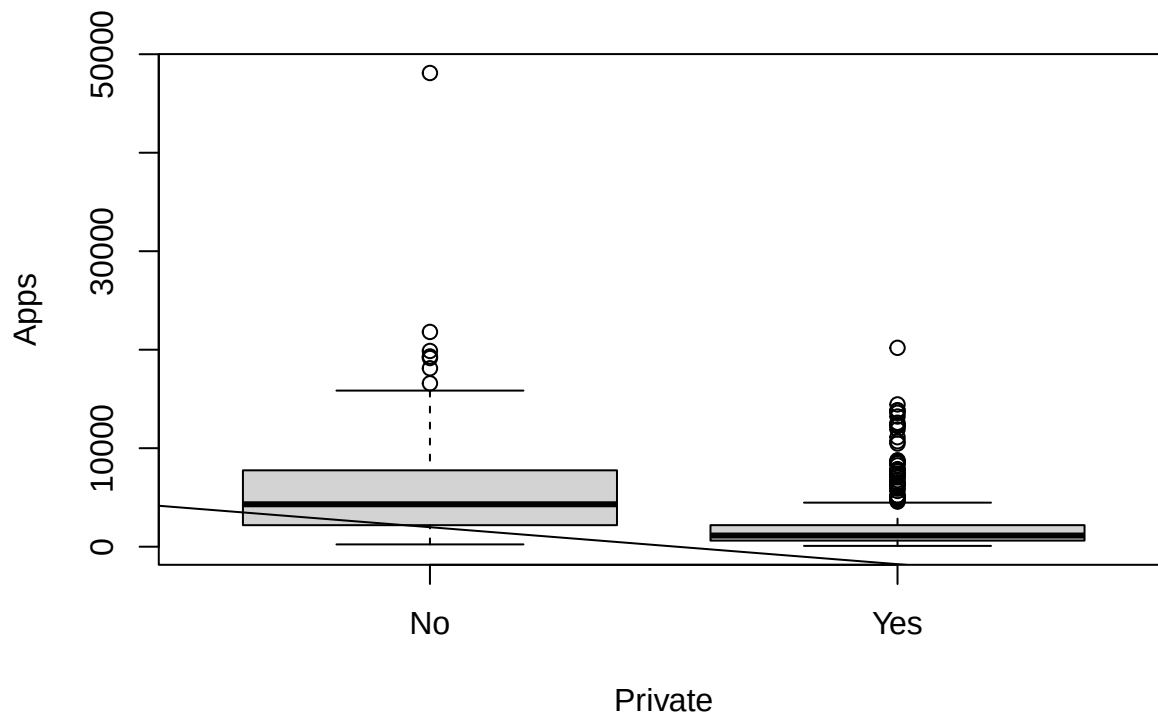
Apply a log or sqrt transformation to the predictor (Expend) to suppress outliers. Another option is to use robust regression where smaller weights are used for outliers.

- Now predict the response Apps by just using the binary variable Private as predictor. How does the regression model look like? Interpret the meaning of the regression coefficients.

```
College_apps_lm <- lm(Apps ~ Private, College)
coef(College_apps_lm)
```

```
## (Intercept) PrivateYes
##      5729.920    -3751.991
```

```
plot(x = College$Private,
     y = College$Apps,
     xlab = "Private",
     ylab = "Apps")
abline(College_apps_lm)
```



The negative regression coefficient for PrivateYes indicates that private schools receive less applications than public schools. R codes no as 0 and yes as 1. For no, the predicted Apps is simply the intercept, β_0 . For yes, the predicted Apps is $\beta_0 + \beta_1$.

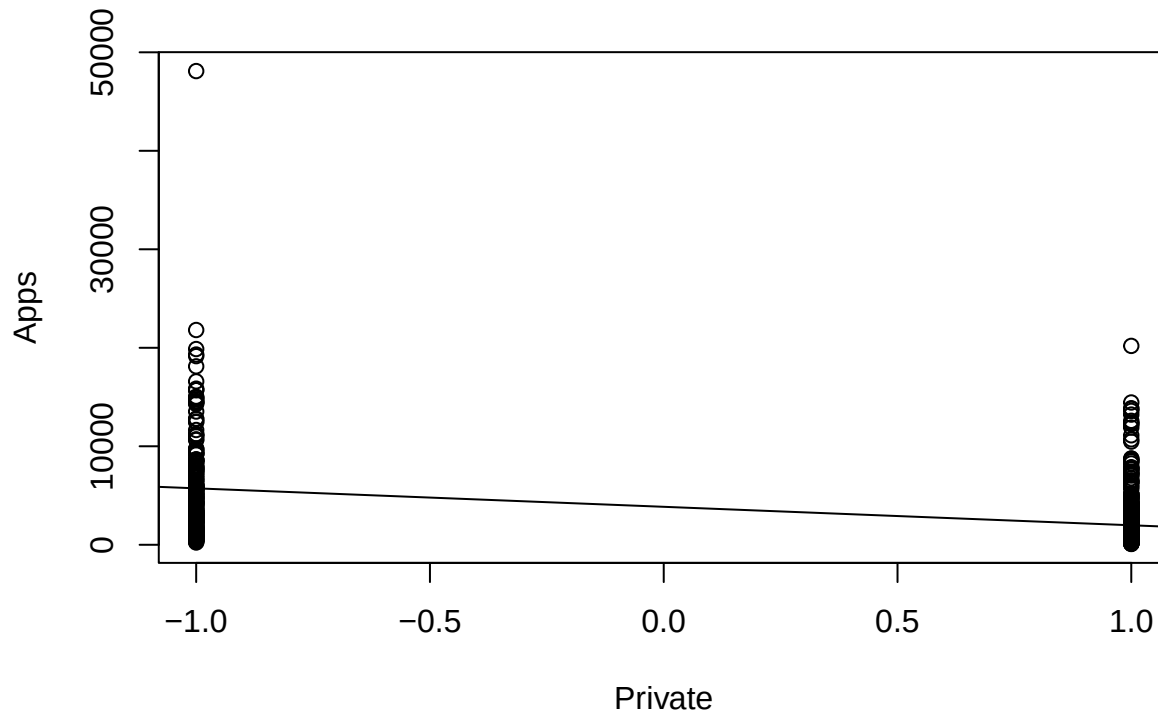
- Convert Private to a variable with levels ± 1 , and regress Apps on this variable. How does the regression model look like? Interpret the meaning of the regression coefficients.

```
College$PrivateNum <- ifelse(College$Private == "Yes", 1, -1)
College_apps_recoded_lm <- lm(Apps ~ PrivateNum, College)
coef(College_apps_recoded_lm)
```

```
## (Intercept) PrivateNum
##      3853.925    -1875.995
```

```
plot(x = College$PrivateNum,
     y = College$Apps,
     xlab = "Private",
```

```
ylab = "Apps")
abline(College_apps_recoded_lm)
```



The intercepts and the coefficients are now different, but the predicted values are the same. The regression equation is $Apps = \beta_0 + \beta_0 \times PrivateNum$. For $PrivateNum = -1$ (no), the predicted *Apps* is 5729, which is equal to the intercept found in 3.

```
# 66% of the sample size
smp_size <- floor(2/3 * nrow(College))

# set the seed to make partition reproducible
set.seed(1)
train_ind <- sample(seq_len(nrow(College)), size = smp_size)

train <- College[train_ind, ]
test <- College[-train_ind, ]

# extract relevant columns for train and test
var_names <- c("Outstate", "Room.Board", "S.F.Ratio", "Expend", "Grad.Rate", "Apps")
train <- train[var_names]
test <- test[var_names]
```

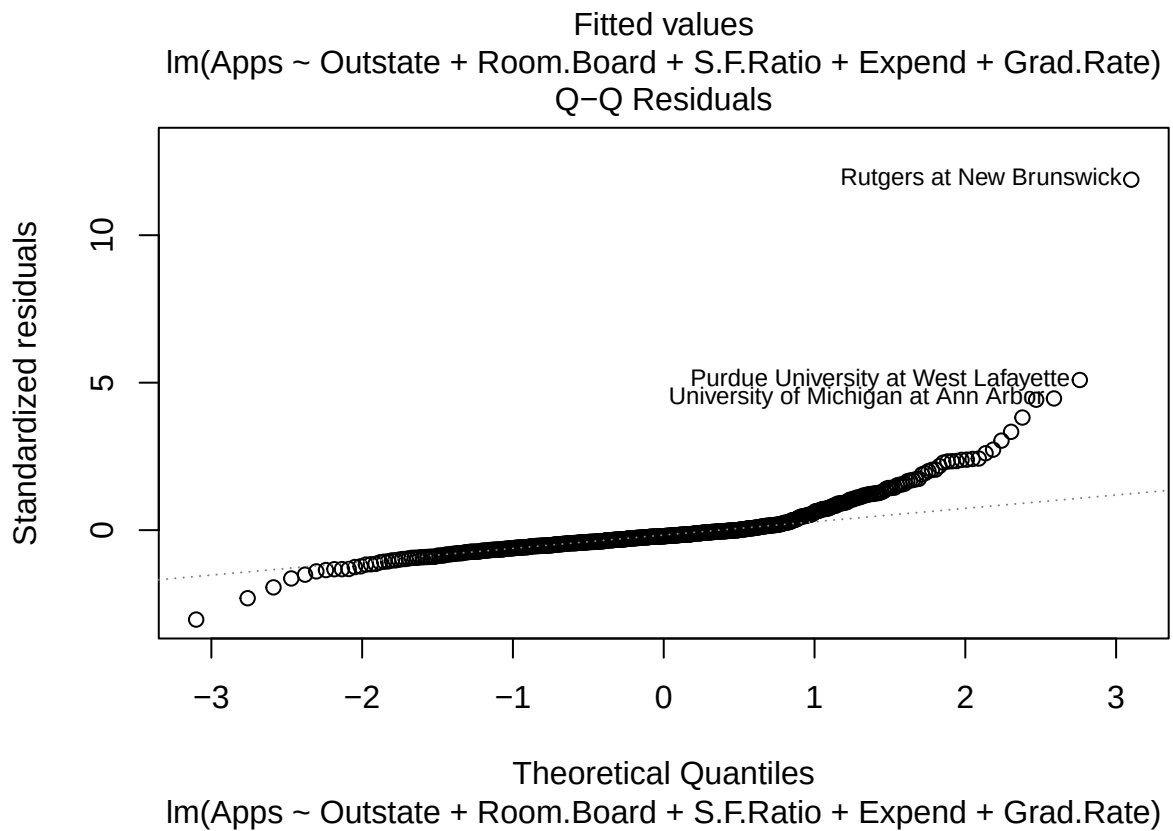
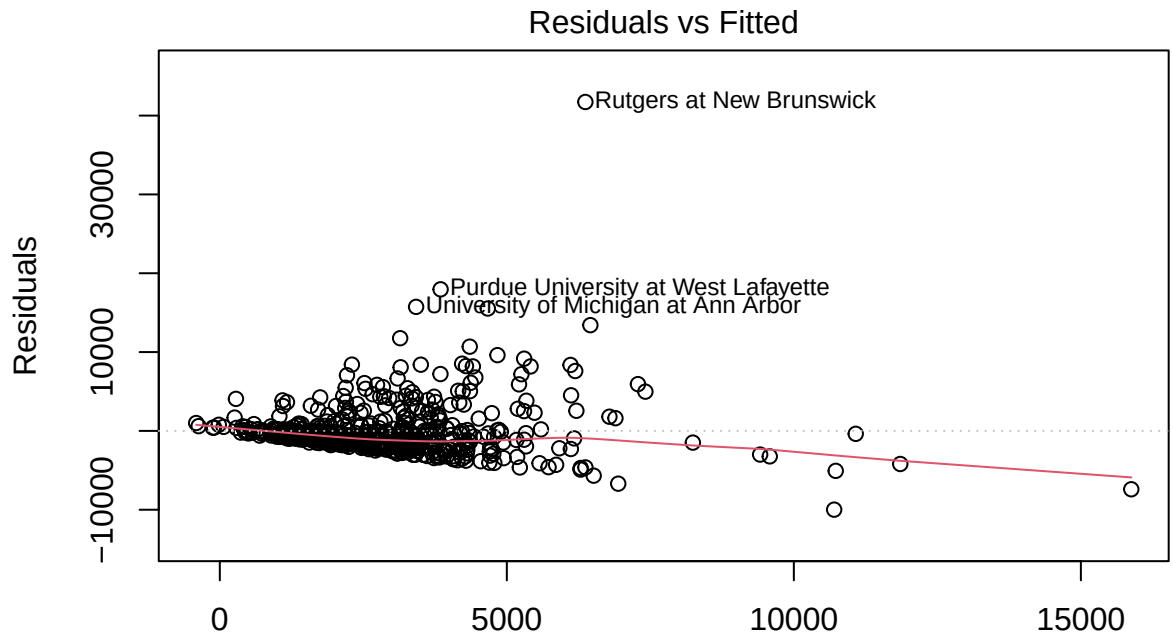
5. Predict the response *Apps* by using all variables in the data frame that make sense content-wise. Inspect and interpret (as far as possible) the diagnostic plots. Are they supporting our model requirements?

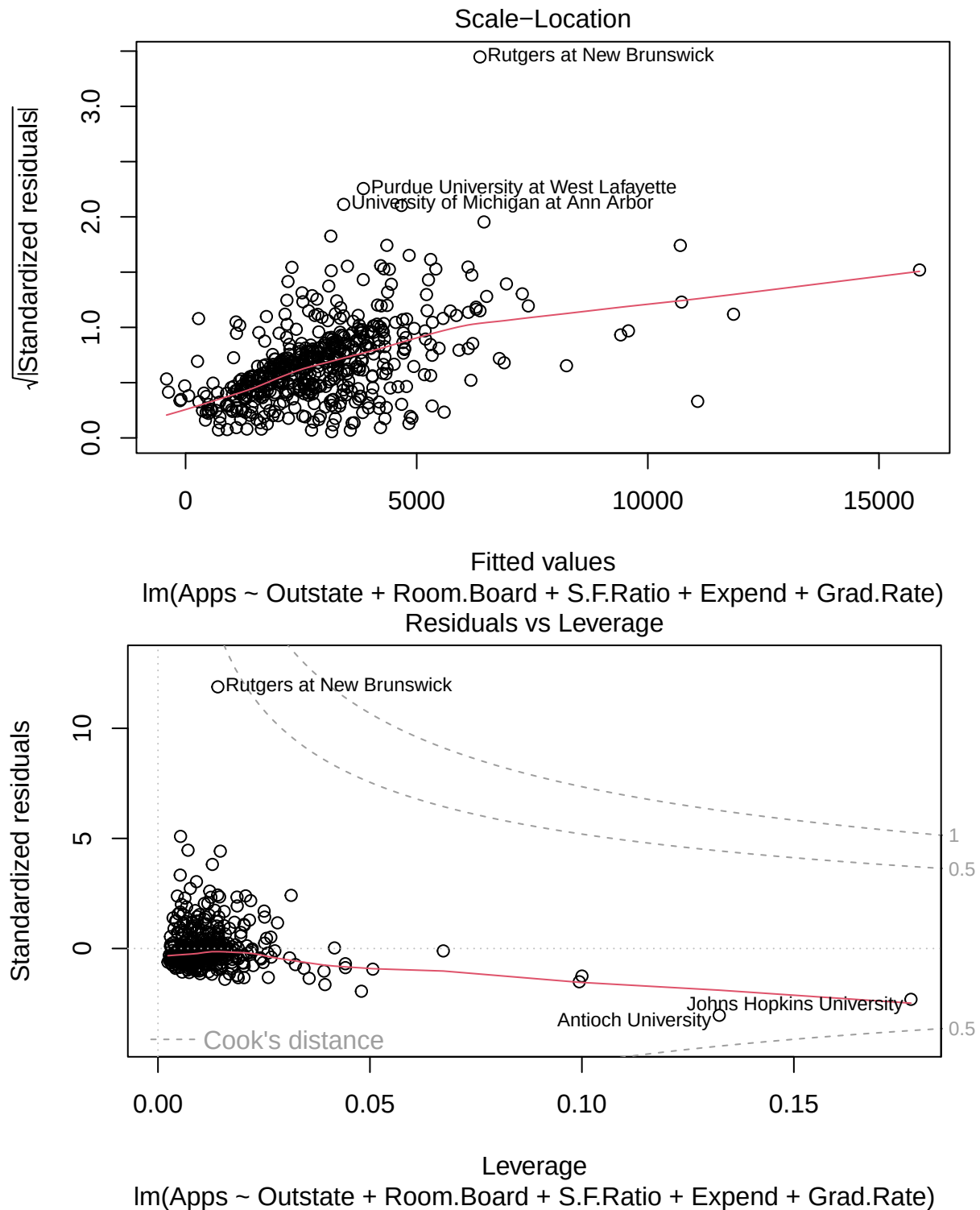
The following regression uses the predictors, *Outstate*, *Room.Board*, *S.F.Ratio*, *Expend*, and *Grad.Rate*.

```
library(ggplot2)

# estimate regression model and predict
train_lm <- lm(Apps ~ Outstate + Room.Board + S.F.Ratio + Expend + Grad.Rate, train)
train_pred <- predict(train_lm, train)
test_pred <- predict(train_lm, test)
```

```
# plotting
plot(train_lm)
```





The residuals vs fitted values plot does not show random scatter around 0 and suggests that the linearity and constant variance requirements are not met.

6. As the regression coefficients depend on the scale of the explanatory variables, we cannot compare them directly. For a comparison, we need to scale the variables to variance 1, which can be done by `scale()`. Estimate the regression model as in 5. based on scaled variables. What can you conclude by inspecting the coefficients?

```

# calculate train mean and sd
train_mean <- sapply(train, mean)
train_sd   <- sapply(train, sd)

# apply scaling
train_scaled <- as.data.frame(scale(train,
                                   center = train_mean,
                                   scale = train_sd))

test_scaled <- as.data.frame(scale(test, # does not have variance of 1
                                   center = train_mean,
                                   scale = train_sd))

# estimate regression model
train_scaled_lm <- lm(Apps ~ Outstate + Room.Board + S.F.Ratio + Expend + Grad.Rate,
                      train_scaled)
train_scaled_pred <- predict(train_scaled_lm, train_scaled)
test_scaled_pred <- predict(train_scaled_lm, test_scaled)

coef(train_scaled_lm)

##      (Intercept)      Outstate      Room.Board      S.F.Ratio      Expend
## -6.479488e-16 -2.893420e-01  2.192402e-01  3.052845e-01  4.936554e-01
##      Grad.Rate
##  1.436019e-01

```

Expend is the most strongly correlated variable, with S.F.Ratio following behind.

7. Compute for models 5. and 6. the RMSEs of training and test set. What do you conclude? Do the models 5. and 6. perform equally well? Do they even lead to the identical predictions? Can we see this based on the RMSEs?

```

library('Metrics')

precision <- 3

# model 5. train RMSE
train_RMSE <- round(rmse(train$Apps, train_pred), precision)
# test RMSE
test_RMSE <- round(rmse(test$Apps, test_pred), precision)

# model 6. train RMSE
train_scaled_RMSE <- round(rmse(train_scaled$Apps, train_scaled_pred), precision)
# test RMSE
test_scaled_RMSE <- round(rmse(test_scaled$Apps, test_scaled_pred), precision)

## [1] "Model 5 train RMSE: 3515.897"
## [1] "Model 5 test RMSE: 3366.552"
## [1] "Model 6 train RMSE: 0.898"
## [1] "Model 6 test RMSE: 0.86"

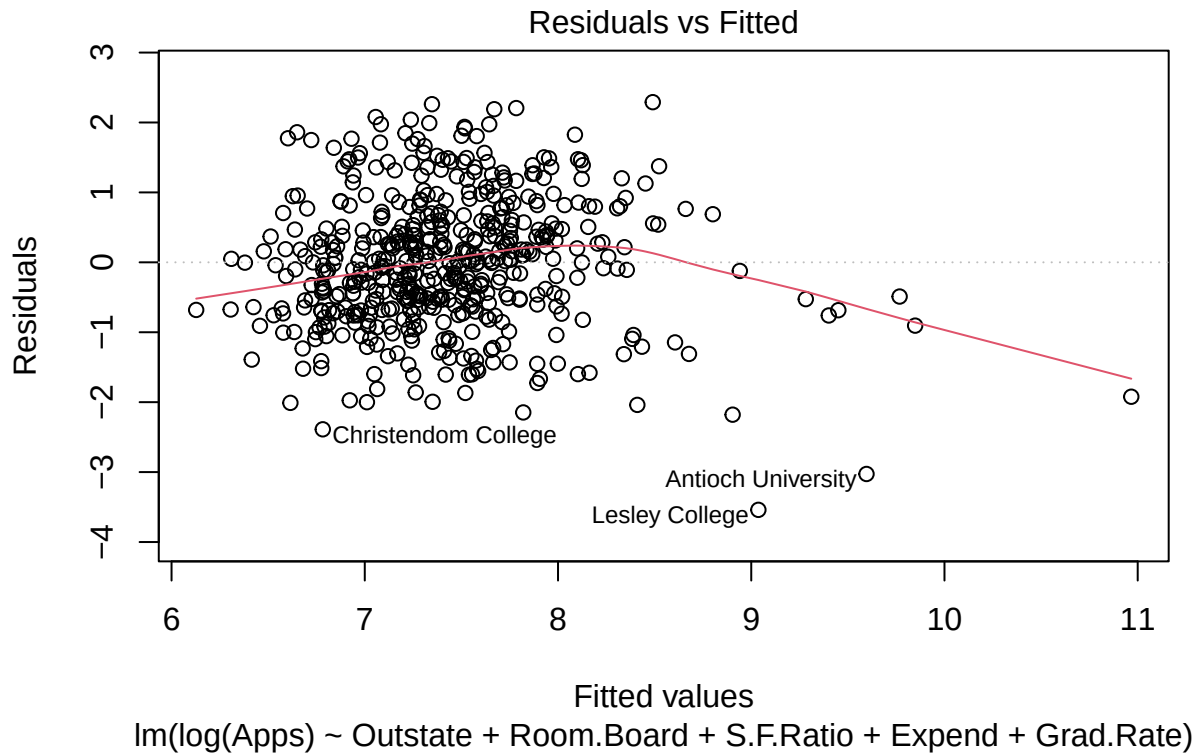
```

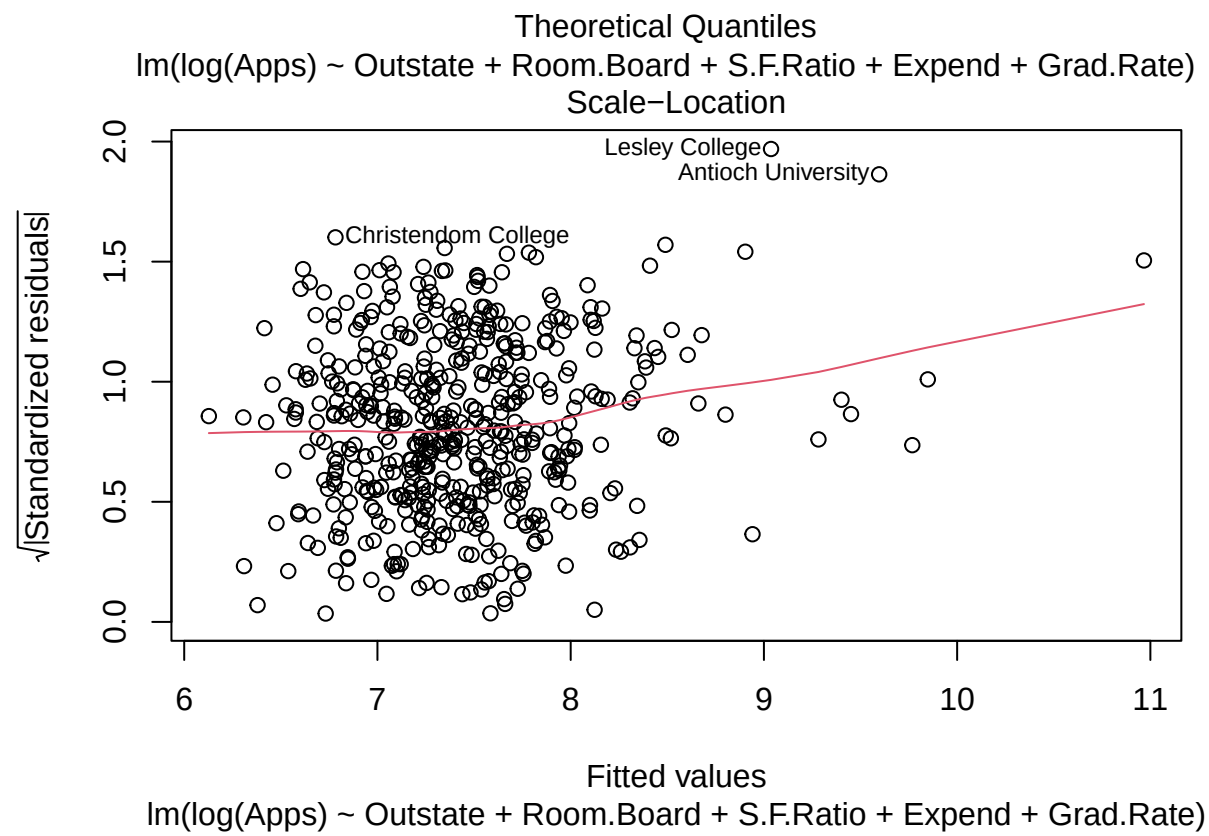
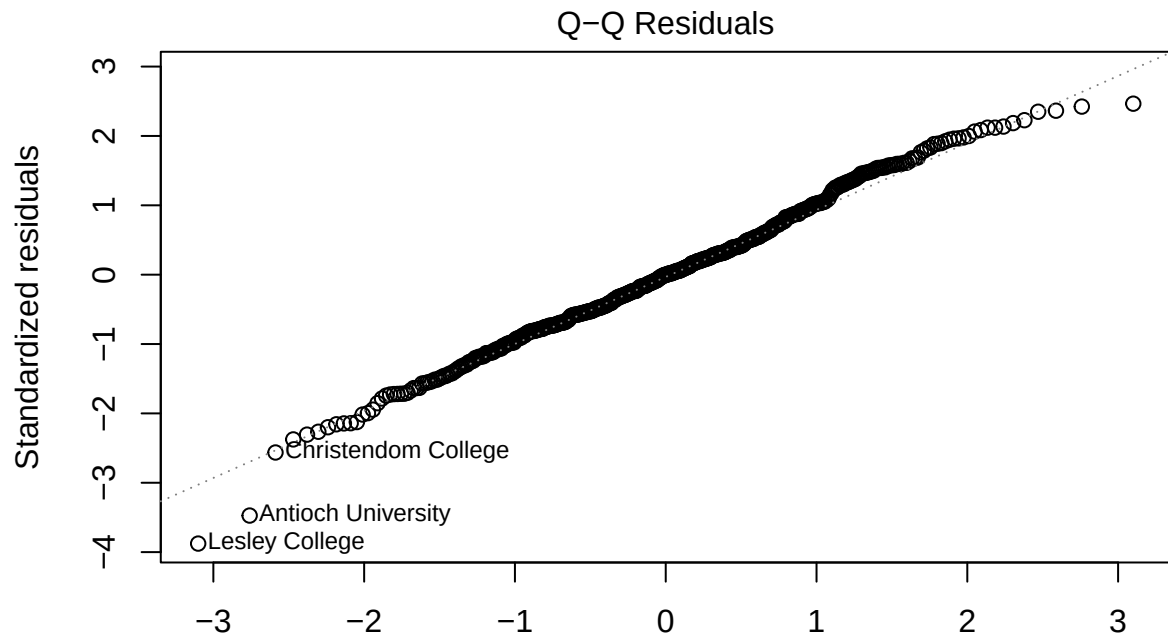
Models 5 and 6 are on different scales, so direct comparisons between the models are not meaningful. However, the results do show that both models perform worse on the train set than the test set. The percent difference between the train and test sets for both models are also similar.

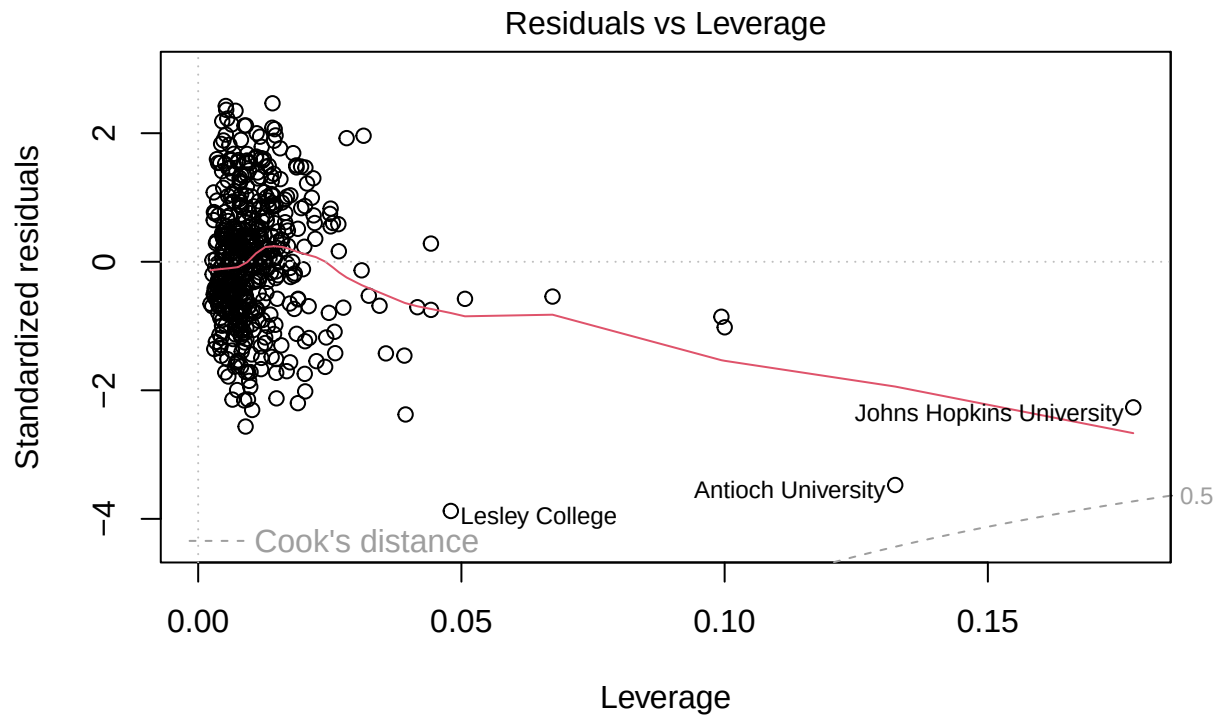
8. The diagnostic plots may suggest that the response could be transformed. Repeat 5. for the log-transformed response. Is the model more appropriate? In which sense?

```
# estimate regression model with transformed response and predict
train_trans_lm <- lm(log(Apps) ~ Outstate + Room.Board + S.F.Ratio + Expend + Grad.Rate, train)
train_trans_pred <- predict(train_trans_lm, train)
test_trans_pred <- predict(train_trans_lm, test)

# plotting
plot(train_trans_lm)
```







The residuals vs fitted plot shows better linearity and more constant variance than model 5. However, some nonlinearity is still present due to the trendline taking a negative slope for larger fitted values.

9. How could we identify if model 5. or 8. is performing better? Obviously, we cannot compare the RMSEs.

Transform the log-transformed model back to the original scale using `exp()`. RMSEs can then be directly compared.